

Lecture 23 Notes: Market Efficiency, WACC, and Capital Structure

Corporate Finance

I. Market Equilibrium and Intrinsic Value

Key Definitions

- **Intrinsic Value ($\hat{P}_0$):** The value estimated based on company fundamentals (cash flows, earnings, management quality, business model).
- **Market Price (P_0):** The current trading price of the stock.
- **Market Equilibrium:** Occurs when intrinsic value equals market price.

Investment Decision Rules

Condition	Action
$\hat{P}_0 > P_0$ (Undervalued)	BUY (price should rise)
$\hat{P}_0 < P_0$ (Overvalued)	SELL (price should fall)
$\hat{P}_0 = P_0$ (Fairly valued)	HOLD

How Prices Move Toward Equilibrium

- Smart investors identify undervalued stocks through fundamental analysis.
- When many investors buy an undervalued stock, demand increases and the price rises toward intrinsic value.
- When a stock is overvalued, selling pressure causes the price to drop toward intrinsic value.
- The speed of this adjustment depends on market efficiency.

II. Market Efficiency

Definition

In an **efficient market**, news is incorporated into the stock price almost instantly.

Forms of Market Efficiency

- **Weak Form:** Past price information is already reflected in current prices.
- **Semi-Strong Form:** All public information is reflected in current prices.
- **Strong Form:** All information (public and private) is reflected in current prices.

Example: Drug Approval News

- If a pharmaceutical company receives FDA approval for a new drug, the news hits the market.
- In efficient markets (U.S., China, India), the stock price adjusts within minutes.
- Digital media makes information flow very fast.
- In less developed markets, lags may still exist.

III. Random Walk Theory

Random Walk Concept

The current stock price already incorporates all available information. Therefore, **tomorrow's price does not depend on today's price.**

Key Implications

- Price changes occur **randomly** because they follow new information.
- Like a coin toss: getting heads today does not affect the probability of heads or tails tomorrow (remains 50/50).
- **Technical analysis** (predicting prices from past trends) is often criticized because stock prices follow a random walk.
- **Fundamental analysis** (analyzing company strength) can identify undervalued stocks, but past price trends cannot predict future movements.

Market Manipulation

- Big players can manipulate the market, especially in less developed markets.
- Illegal "pooling" and position building still occur.
- However, the market ultimately responds to supply and demand.

IV. Dividends and Share Repurchases

Why Do Companies Pay Dividends?

Dividends are "sticky" - once a company pays a dividend, shareholders expect it to continue.

Dividend Signaling

- If a company pays a \$5 dividend this year, shareholders expect the same or higher next year.
- If a company stops paying dividends, it sends a **negative signal** that the company lacks funds.

Share Repurchases (Buybacks)

- Some companies use share repurchases to return cash to shareholders.
- Buybacks allow companies to return cash at market price without committing to a permanent dividend increase.
- Common in both U.S. and Pakistani markets.

V. Components of Capital and WACC

Three Main Components of Capital

1. **Debt:** Use after-tax cost of debt
2. **Preferred Stock:** Preferred dividend divided by price
3. **Equity:** Comes from retained earnings OR issuing new shares

Cost of Equity Sources

- **Retained Earnings:** Earnings that belong to shareholders but are kept in the company.
- **New Share Issuance:** Raising capital by issuing additional common stock.

WACC Formula

$$\text{WACC} = w_d \times r_d(1 - T) + w_p \times r_p + w_s \times r_s$$

Where:

- w_d, w_p, w_s = Weights of debt, preferred stock, and equity
- r_d = Before-tax cost of debt

- T = Marginal tax rate
- r_p = Cost of preferred stock = D_p/P_p
- r_s = Cost of common equity

Market Value Weights vs. Book Value Weights

Important Note

When calculating WACC, you should ideally use **market value weights** rather than book value weights.

$$\text{Market Value of Equity} = \text{Shares Outstanding} \times \text{Current Market Price}$$

VI. Financial Statements: Net Income to EPS

Earnings Per Share (EPS) Calculation

$$\text{EPS} = \frac{\text{Earnings Available for Common Shareholders}}{\text{Shares Outstanding}}$$

Income Statement Flow

Net Income $\rightarrow$ Subtract Preferred Dividends $\rightarrow$ Earnings Available for Common Shareholders $\rightarrow$ Dividends

Common Mistake to Avoid

Do NOT use Net Income directly in EPS calculation. You must first subtract preferred dividends.

Example:

$$\begin{aligned} \text{Net Income} &= \$10,000,000 \\ \text{Preferred Dividends} &= \$2,000,000 \\ \text{Earnings Available for Common} &= \$10,000,000 - \$2,000,000 = \$8,000,000 \\ \text{Shares Outstanding} &= 4,000,000 \\ \text{EPS} &= \frac{\$8,000,000}{4,000,000} = \$2.00 \text{ per share} \end{aligned}$$

VII. Flotation Costs and New Equity

Definition

Flotation costs are costs incurred when issuing new shares, including underpricing, investment banker fees, legal fees, and prospectus costs.

Effect on Cost of Equity

When new shares are issued, the company receives less than the market price due to flotation costs.

$$\text{Net Proceeds per Share} = P_0 - \text{Flotation Cost per Share}$$

Example

$$P_0 = \$40 \text{ (market price)}$$

$$D_0 = \$2.00$$

$$g = 6\% = 0.06$$

$$\text{Flotation Cost} = \$4 \text{ per share}$$

$$\text{Net Proceeds} = \$40 - \$4 = \$36$$

Cost of Retained Earnings:

$$r_s = \frac{D_0(1+g)}{P_0} + g = \frac{2.00 \times 1.06}{40} + 0.06 = \frac{2.12}{40} + 0.06 = 0.053 + 0.06 = 11.3\%$$

Cost of New Equity:

$$r_s = \frac{D_0(1+g)}{\text{Net Proceeds}} + g = \frac{2.12}{36} + 0.06 = 0.0589 + 0.06 = 11.89\%$$

Key Insight

Flotation costs **increase the cost of capital** for the firm because the company receives less net cash for the same dividend obligation.

VIII. Pecking Order Theory

Ideal Sequence for Raising Funds

A company should **never issue new shares as its first choice**. The pecking order is:

1. **Internal Funds (Retained Earnings)** - First choice, no flotation costs
2. **Debt** - Second choice, provides tax shield
3. **New Equity** - Last resort, most expensive due to flotation costs

WACC Problem-Solving Tip

When solving WACC questions:

- If both cost of retained earnings and cost of new shares are given, use the **cost of retained earnings** unless the problem specifically states that new equity must be issued.
- New equity is always more expensive due to flotation costs.

IX. Summary of Key Formulas

Formula Summary

$$\text{WACC : } \text{WACC} = w_d r_d (1 - T) + w_p r_p + w_s r_s \quad (1)$$

$$\text{Cost of Preferred Stock : } r_p = \frac{D_p}{P_p} \quad (2)$$

$$\text{Cost of Retained Earnings : } r_s = \frac{D_1}{P_0} + g \quad (3)$$

$$\text{Cost of New Equity : } r_s = \frac{D_1}{P_0 - F} + g \quad (4)$$

$$\text{EPS : } \text{EPS} = \frac{\text{Net Income} - \text{Preferred Dividends}}{\text{Shares Outstanding}} \quad (5)$$

$$\text{Market Value of Equity : } \text{Market Value} = \text{Shares} \times P_0 \quad (6)$$

X. Key Terms Glossary

Term	Definition
Intrinsic Value	Estimated true value based on fundamentals
Market Equilibrium	When intrinsic value equals market price
Efficient Market	News is incorporated into prices almost instantly
Random Walk	Future prices cannot be predicted from past prices
Flotation Costs	Costs incurred when issuing new securities
Pecking Order	Preferred sequence: retained earnings → debt → new equity
Market Value Weights	Weights based on current market prices (preferred for WACC)